RANDOM FOREST MACHINE LEARNING PROJECT

Heart Disease Classification

Dataset: heart.csv (https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction)

Records: 918

Columns: 12

Target: HeartDisease (0/1)

Problem Statement:

Heart disease is one of the major health concerns worldwide, and early identification can help in better medical decision-making. Traditional diagnosis involves analyzing several health-related factors, which can be time-consuming. This project aims to develop a machine learning model using the Random Forest algorithm to predict the possibility of heart disease based on patient health attributes. The model analyzes the dataset, identifies important features, and classifies individuals into heart disease and no heart disease categories.

Data Preprocessing:

The dataset was explored and checked for missing values. Categorical variables were converted into numerical values using one-hot encoding. The target column was separated, and the dataset was divided into 80% training and 20% testing data.

Model Used:

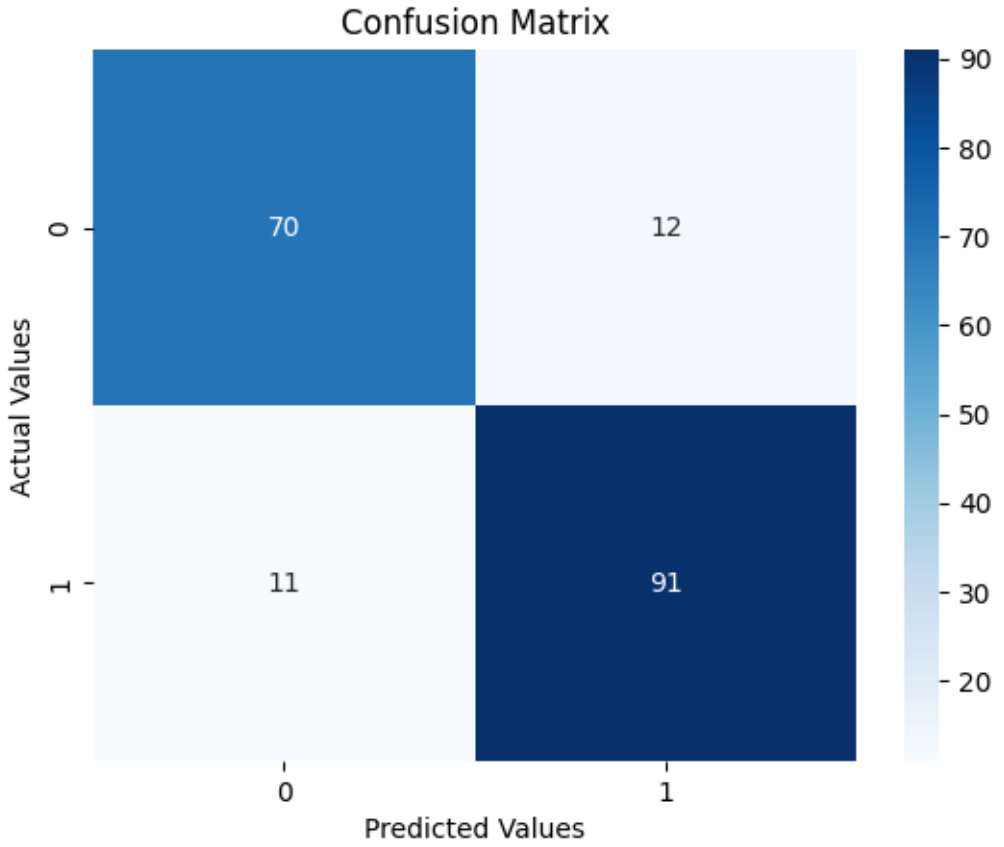
The Random Forest algorithm was applied to build a classification model. The model was trained using the training dataset to predict heart disease and classify the results into two categories: 0 and 1.

Evaluation Metrics:

Metric	Test result
Accuracy	87.50%
Precision	88.35%
Recall	89.22%
F1-score	88.78%

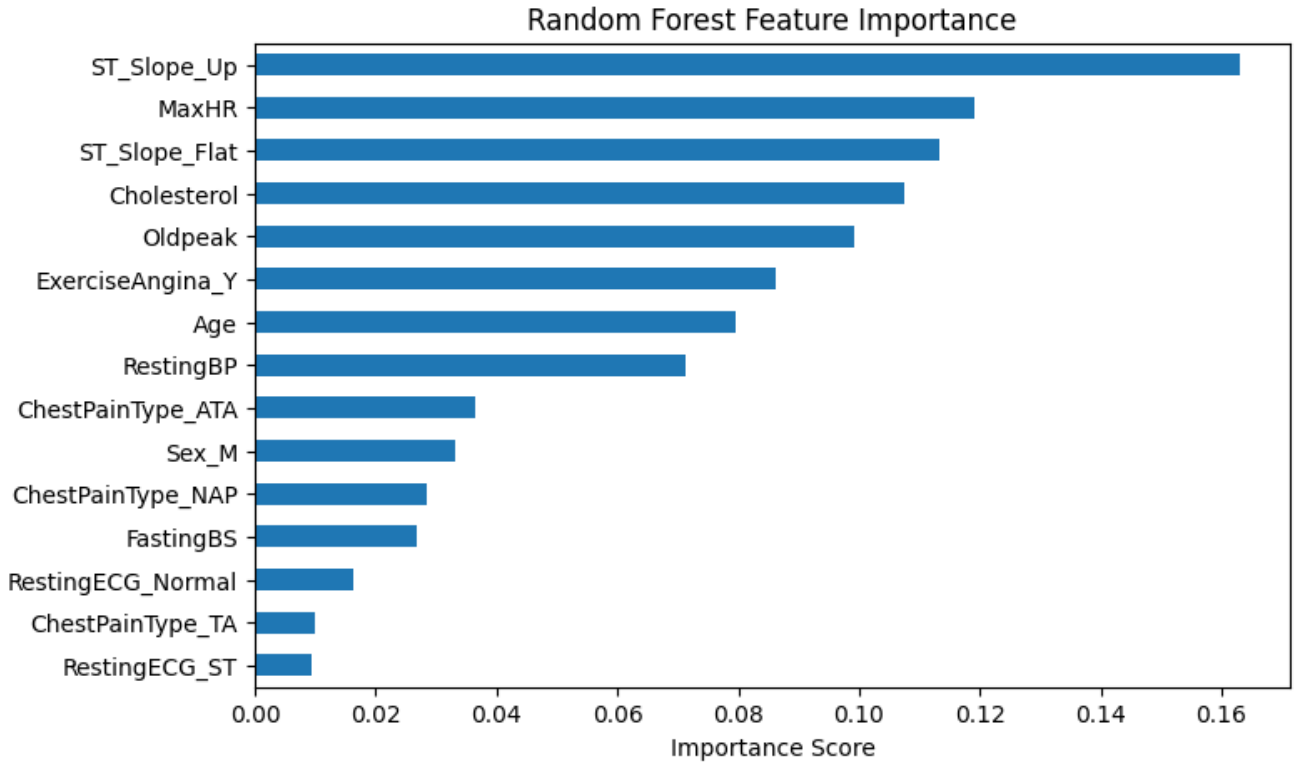
Confusion Matrix:

The confusion matrix shows the model's correct and incorrect predictions. It correctly classified 70 records without heart disease and 91 records with heart disease, while 23 records were incorrectly classified.



Feature Importance:

Feature importance is used to identify which input attributes contribute most to the Random Forest model's predictions. In this project, ST_Slope_Up is the most important feature, followed by MaxHR, ST_Slope_Flat, Cholesterol, and Oldpeak. The graph displays the importance scores of all features, helping us understand which attributes have a greater influence on the model's heart disease classification.



Key Findings:

- The Random Forest model achieved an accuracy of 87.50% on the test dataset.
- Precision, recall, and F1-score were 88.35%, 89.22%, and 88.78%, respectively.
- The confusion matrix shows that 161 out of 184 records were correctly classified.
- ST_Slope_Up was identified as the most important feature, followed by MaxHR and ST_Slope_Flat.
- The model successfully classified patients into heart disease and no heart disease categories.

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Conclusion:

The Random Forest algorithm was successfully implemented on the Heart Disease dataset to predict the presence or absence of heart disease. The model achieved 87.50% accuracy with good precision, recall, and F1-score. Feature importance analysis helped identify the key attributes influencing the predictions. Overall, the project demonstrates the effectiveness of Random Forest in performing heart disease classification using machine learning.

Kaggle source: https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

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